$\vec{B} = B_0 \hat{z}$, $\vec{k} = k\hat{z}$, $\vec{E}_1 = E_x \hat{x} + E_y \hat{y}$

动力学方程：

$$m\frac{d\vec{v}}{dt} = -e(\vec{E} + \vec{v}_1 \times \vec{B}_0) ,$$

$$-im\omega \vec{v}_{e1} = -e(\vec{E} + \vec{v}_{e1} \times \vec{B}_0)$$

$$\begin{cases} v_x = \frac{-ie}{m\omega}(E_x + v_y B_0) \\ v_y = \frac{-ie}{m\omega}(E_y - v_x B_0) \end{cases}$$

$$\Rightarrow \quad v_x = -\frac{ie}{m\omega}\left[E_x - \frac{ieB_0}{m\omega}\left(\frac{-ieB_0}{m\omega}E_y + \frac{ieB_0}{m\omega}v_x\right)\right]$$

$$= -\frac{ie}{m\omega}E_x - \frac{e^2B_0^2}{m^2\omega^2}E_y + \frac{e^2B_0^2}{m^2\omega^2}v_x$$

$$\therefore \quad v_x = \frac{e}{m\omega}\left(-iE_x - \frac{\omega_c}{\omega}E_y\right)\left(1 - \frac{\omega_c^2}{\omega^2}\right)^{-1}$$

$$v_y = \frac{e}{m\omega}\left(-iE_y + \frac{\omega_c}{\omega}E_x\right)\left(1 - \frac{\omega_c^2}{\omega^2}\right)^{-1}.$$

$$\omega_c = \frac{eB_0}{m}.$$

consider

$$-\vec{k}(\vec{k}\cdot\vec{E}) + k^2\vec{E}_1 = i\omega\mu_0\vec{j}_1 + \frac{\omega^2}{c^2}\vec{E}_1$$

$$(\omega^2 - c^2k^2)\vec{E}_1 = -i\omega\mu_0 c^2\vec{j}_1 = i\omega\mu_0 c^2 n_0 e\vec{v}_{e1}$$

分量形式：

$$\begin{cases} (\omega^2 - c^2k^2)E_x = i\omega\mu_0 c^2 n_0 e\dfrac{e}{m\omega}\left(-iE_x - \dfrac{\omega_c}{\omega}E_y\right)\left(1-\dfrac{\omega_c^2}{\omega^2}\right)^{-1} \\ (\omega^2 - c^2k^2)E_y = i\omega\mu_0 c^2 ne\dfrac{e}{m\omega}\left(-iE_y + \dfrac{\omega_c}{\omega}E_x\right)\left(1-\dfrac{\omega_c^2}{\omega^2}\right)^{-1} \end{cases}$$

ω_p^2

$$\alpha = \frac{\omega_p^2}{1-\omega_c^2/\omega^2}$$

$$\therefore \quad \begin{cases} (\omega^2 - c^2k^2 - \alpha)E_x + i\alpha\dfrac{\omega_c}{\omega}E_y = 0 \\ (\omega^2 - c^2k^2 - \alpha)E_y - i\alpha\dfrac{\omega_c}{\omega}E_x = 0 \end{cases}$$

$$\det|A| = \begin{vmatrix} (\omega^2 - c^2k^2 - \alpha) & i\alpha\dfrac{\omega_c}{\omega} \\ -i\alpha\dfrac{\omega_c}{\omega} & (\omega^2 - c^2k^2 - \alpha) \end{vmatrix} = 0$$

$$\Rightarrow \quad \omega^2 - c^2k^2 = \alpha\left(1 \pm \frac{\omega_c}{\omega}\right)$$

$$= \frac{\omega_p^2}{1 \mp \dfrac{\omega_c}{\omega}}$$

色散关系：

$$\frac{c^2k^2}{\omega^2} = 1 - \frac{\omega_p^2/\omega^2}{1-\omega_c/\omega} \qquad (R\text{波})$$

$$\frac{c^2k^2}{\omega^2} = 1 - \frac{\omega_p^2/\omega^2}{1+\omega_c/\omega} \qquad (L\text{波})$$

$$(\omega^2 - c^2k^2 - \alpha) E_x + i\alpha \frac{\omega_c}{\omega} E_y = 0$$

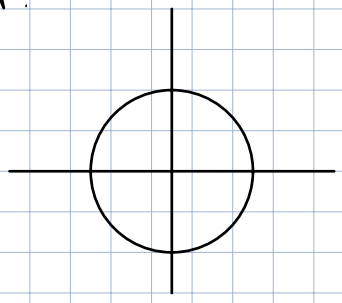
$$(\omega^2 - c^2k^2 - \alpha) E_y - i\alpha \frac{\omega_c}{\omega} E_x = 0$$

$$\Rightarrow \quad \frac{E_x}{E_y} = -\frac{E_y}{E_x}$$

$$E_x^2 + E_y^2 = 0.$$

R波：　$E_x + iE_y = 0.$　　$E_y = iE_x.$

$$\phi_y = \phi_x + \frac{\pi}{2}.$$



L波.　$E_x = iE_y$,　　$\phi_x = \phi_y + \frac{\pi}{2}.$

令 $k=0$ 截止.

R 波: $1-\frac{\omega_c}{\omega}-\frac{\omega_p^2}{\omega^2}=0$

$\Leftrightarrow \quad \omega^2-\omega_c\omega-\omega_p^2=0$

$$\omega_R=\frac{\omega_c+\sqrt{\omega_c^2+4\omega_p^2}}{2}$$ 舍去负解

右旋截止

同理, L 波.

$\omega^2+\omega\omega_c-\omega_p^2=0$

$$\omega_L=\frac{-\omega_c+\sqrt{\omega_c^2+4\omega_p^2}}{2}$$

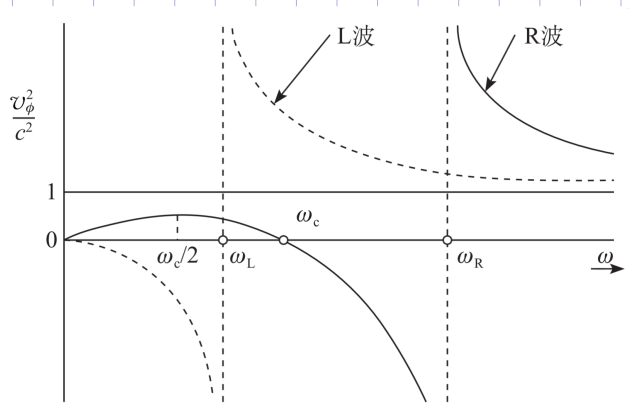
左旋截止.

$\frac{v_p^2}{c^2}$ 与 ω 关系? $v_p=\frac{\omega}{k}$.

R 波 $\frac{v_p^2}{c^2}=\left(1-\frac{\omega_p^2/\omega^2}{1-\omega_c/\omega}\right)^{-1}$ ω_R.

$=\frac{\omega(\omega-\omega_c)}{\omega^2-\omega_c\omega-\omega_p^2}$

- $\omega\to\infty$. $\frac{v_p^2}{c^2}\to 1$. 分母 $=0$. ω_R.
- $\omega=0$ $\frac{v_p^2}{c^2}=0$.
- $\omega=\omega_c$ $v_p^2/c^2=0$.
- $\omega=\frac{\omega_c}{2}$ 极大值.



L波.

$$v_p^2/c^2 = \left(1 - \frac{\omega_p^2/\omega^2}{1+\omega/\omega_c}\right)^{-1}$$

ω_L

- $\omega \to \infty$. $v_p^2/c^2 \to 1$
- $\omega = \omega_L$ $v_p^2/c^2 \to \infty$.

  - $\omega_L < \omega_c$

法拉第旋转.

色散关系：

$$\frac{c^2k^2}{\omega^2} = 1 - \frac{\omega_p^2/\omega^2}{1-\omega_c/\omega} \qquad (R\text{波})$$

$$\frac{c^2k^2}{\omega^2} = 1 - \frac{\omega_p^2/\omega^2}{1+\omega_c/\omega} \qquad (L\text{波})$$

平波波 = R波 + L波.

二者相速度不同. 开始同相，后续不同，产生偏振的旋转.

$$k_R = \frac{\omega}{c}\sqrt{1-\frac{\omega_p^2}{\omega^2(1-\omega_c/\omega)}} \qquad k_L = \frac{\omega}{c}\sqrt{1-\frac{\omega_p^2}{\omega^2(1+\omega_c/\omega)}}$$

$$E = \vec{E}_R e^{i(k_R L-\omega t)} + \vec{E}_L e^{i(k_L L-\omega t)}$$

$E_x + iE_y = 0.$

$$\begin{cases} \vec{E}_R = E_0 e^{i(k_R L-\omega t)}\hat{e}_x + iE_0 e^{i(k_R L-\omega t)}\hat{e}_y \\ \vec{E}_L = E_0 e^{i(k_L L-\omega t)}\hat{e}_x + iE_0 e^{i(k_L L-\omega t)}\hat{e}_y \end{cases}$$

$$\vec{E} = \vec{E}_R + \vec{E}_L = \underbrace{E_0\hat{e}_x[e^{i(k_R L-\omega t)} + e^{i(k_L L-\omega t)}]}_{E_x} + \underbrace{iE_0[e^{i(k_R L-\omega t)} + e^{i(k_L L-\omega t)}]}_{E_y}\hat{e}_y$$

E_x 与 E_y 的相位关系？ 作比值.

$$\frac{E_x}{E_y} = \frac{e^{ik_R L} + e^{ik_L L}}{i(e^{ik_R L} - e^{ik_L L})} = -i\frac{e^{i(k_R L - k_L L)} + 1}{e^{i(k_R L - k_L L)} - 1}$$

形式上像 cot 函数.

考虑 $\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$ $\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2}$

$$\Rightarrow \cot\theta = -i\frac{e^{i2\theta} + 1}{e^{-i2\theta} - 1}, \quad \therefore \theta = -\frac{k_R L - k_L L}{2} = \frac{k_L L - k_R L}{2}$$

线偏光

又 $$k_L - k_R = \frac{\omega}{c}\left(\sqrt{1 - \frac{\omega_p^2}{\omega^2(1 + \frac{\omega_c}{\omega})}} - \sqrt{1 - \frac{\omega_p^2}{\omega^2(1 - \frac{\omega_c}{\omega})}}\right).$$

由于 $\omega \gg \omega_p \gg \omega_c$，作 Taylor 展开：

$$k_L - k_R = \frac{1}{2}\frac{\omega}{c}\left[\frac{\omega_p^2}{\omega^2(1 - \frac{\omega_c}{\omega})} - \frac{\omega_p^2}{\omega^2(1 + \frac{\omega_c}{\omega})}\right]$$

$$= \frac{1}{c}\frac{\omega_p^2 \omega_c}{\omega^2(1 - \omega_c/\omega)(1 + \omega_c/\omega)}$$

$$\approx \frac{\omega_p^2 \omega_c}{c\omega^2} = \frac{n_0 e^3 B_0}{c\varepsilon_0 m^2 \omega^2}$$